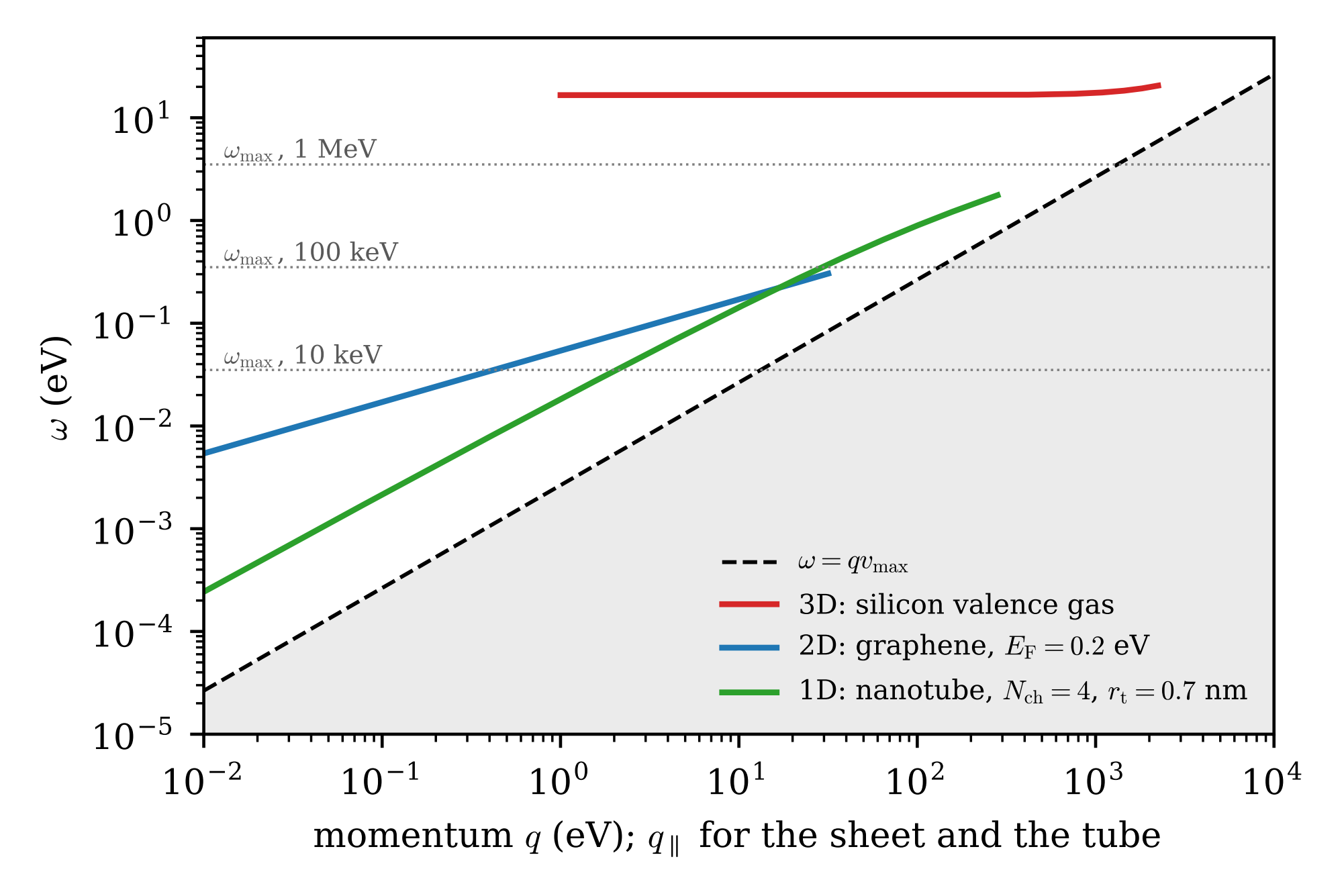
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
$\omega_{\max}$, 1 MeV
$\omega_{\max}$, 100 keV
$\omega_{\max}$, 10 keV
ω (eV)
$\omega = qv_{\max}$
3D: silicon valence gas
2D: graphene, $E_{\mathrm{F}} = 0.2$ eV
1D: nanotube, $N_{\mathrm{ch}} = 4$, $r_{\mathrm{t}} = 0.7$ nm
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
momentum q (eV); $q_{\parallel}$ for the sheet and the tube